From reference paper "Study on ride comfort of tractor with tandem suspension based multi body system dynamics."

Mathematical model of vehicle and dynamics eq. which gave a set of (1-8) eqns

But there are for 6X4 tandem
But ours is 6X2 NRS tandem rear axle
has a equaliser pitch.

Rear axle

R1 R2

F Axle

NRS : Non reactive ⇒ no equalising beam, NO load transfer (no dynamic load sharing)

⇒ 2 axle move independently

So Now the modified set of equations from that paper.

Eq-① Seat vertical motion.

$$M_{ch}\ddot{z}_{ch} + k_{ch}(z_{ch} - (z_c - e\theta))$$

$$+ c_{ch}(\dot{z}_{ch} - (\dot{z}_c - e\dot{\theta})) = 0$$

Eq-2 Cabin - driving room.

$$M_{dr}\ddot{z}_{dr} + k_{dr}(z_{dr} - (z_c - e\theta))$$

$$+ c_{dr}(\dot{z}_{dr} - (\dot{z}_c - e\dot{\theta})) = 0$$

chassis eqn quite changed from paper:
As there was a reactive model with middle & rear suspension force enter through a equaliser node $z_3$

Heave
Eq-3

$$M_2 \ddot{z}_c + k_f (z_c - a\theta - z_{1f}) + c_f(\dot{z}_c - a\dot{\theta} - \dot{z}_{1f})$$
$$+ k_m (z_c + b_m \theta - z_{1m})$$
$$+ c_m (z_c + b_m \dot{\theta} - \dot{z}_{1m})$$
$$+ k_r (z_c + b_r \theta - z_{1r})$$
$$+ c_r (\dot{z}_c + b_r \dot{\theta} - \dot{z}_{1r}) = 0$$

chassis pitch $\quad I_{yy} \sim M_2 q_{2y}^2$

Eq-4

$$I_{yy} \ddot{\theta} - a[k_f (z_c - a\theta - z_{1f}) + c_f (\dot{z}_c - a\dot{\theta} - \dot{z}_{1f})]$$
$$+ b_m [k_m (z_c + b_m \theta - z_{1m}) + c_m (\dot{z}_c + b_m \dot{\theta} - \dot{z}_{1m})]$$
$$+ b_r [k_r (z_c + b_r \theta - z_{1r})] + c_r (\dot{z}_c + b_r \dot{\theta} - \dot{z}_{1r})] = 0$$

∴ Each rear axle acts independently on chassis - NRS - rear suspension

Axles eqn:-

$F_1$
Eq-5.

$$M_{1f} \ddot{z}_{1f} + k_f (z_{1f} - (z_c - a\theta))$$
$$+ c_f (\dot{z}_{1f} - (\dot{z}_c - a\dot{\theta})) + k_{tf} (z_{1f} - q_f) = 0$$

Here again as outer NRS tandem no equaliser node

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$\gamma_1$

Eq 6. $M_{1m}\ddot{z}_{1m} + k_m(z_{1m} - (z_c + b_m\theta))$
$+ c_m(\dot{z}_{1m} - (\dot{z}_c + b_m\dot{\theta})) + k_{tm}(z_{1m} - q_m) = 0$

$\gamma_2$

Eq. 7 $M_{1r}\ddot{z}_{1r} + k_r(z_{1r} - (z_c + b_r\theta))$

$+ c_r(\dot{z}_{1r} - (\dot{z}_c + b_r\dot{\theta})) + k_{tr}(z_{1r} - q_r) = 0$

Shock absorber : same as the paper.

Eq-8 $F_s = -c\dot{\gamma} + k(\gamma - L) + F_{preload}$, $\gamma \neq L$

$F_s =$ preload. $\gamma = L$

Now moving on things which we need to incorporate But were not in paper:

Eq-9 $I_{xx}\ddot{\phi} + \sum_{i \in f,m,r} (k_{\phi,i}\phi + c_{\phi,i}\dot{\phi})$

9(i)
9(ii) $+ \sum_i F_{y,i}\, h_{rc,i} = 0$.
9(iii)

Lateral & yaw dynamics:-

Eq. 10 $m(\dot{v}_y + ur) = F_{y,f} + F_{ym} + F_{yr}$

11 $I_{zz}\dot{r} = aF_{y,f} - b_m F_{y,m} - b_r F_{y,r}$

12 $\dot{\psi} = r$

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Tire slip angles

12(i) $\alpha_f = \delta - \tan^{-1}\left(\frac{v_y + a r}{u}\right)$

(ii) $\alpha_m = -\tan^{-1}\left(\frac{v_y - b_m r}{u}\right)$

(iii) $\alpha_r = -\tan^{-1}\left(\frac{v_y - b_r r}{u}\right)$

Tire forces → Brush/ piece wise linear
$i \rightarrow (f, m, r)$

14. $F_{yi} = \begin{cases} C_{\alpha i}\, \alpha_i & |\alpha_i| \le \alpha_{crit,i} \\ \mu F_{zi}\, \mathrm{sgn}(\alpha_i) & |\alpha_i| > \alpha_{crit,i} \end{cases}$

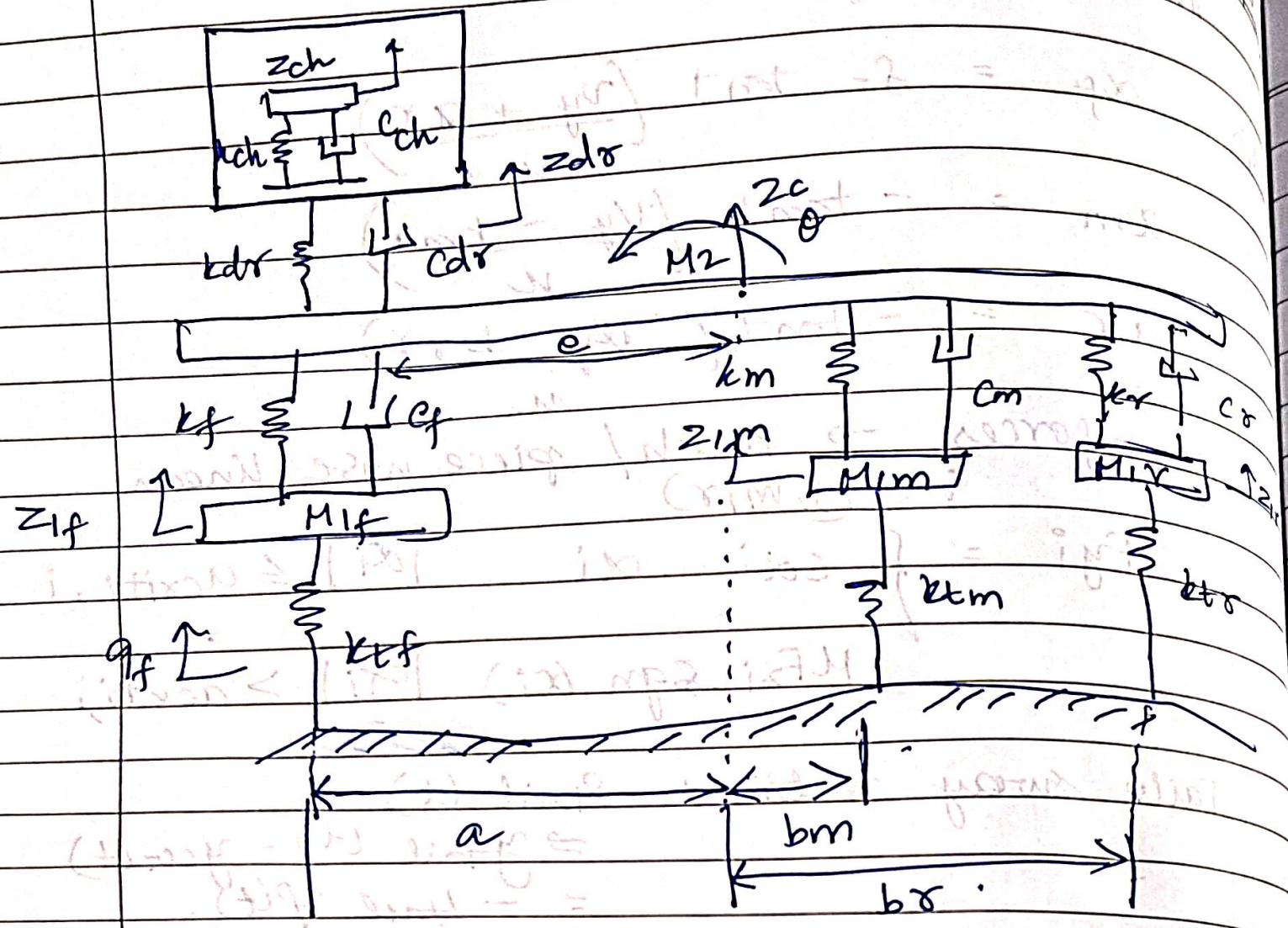
Tail sway metric: $S_{tail}(t)$
$\approx y_{tail}(t) - y_{cg}(t)$
$= -L_{tail}\, \psi(t)$.

Nomenclature:

heave → $z_c$ | lateral velocity → $v_y$
pitch → $\theta$ | yaw rate → $r$
roll → $\phi$ | yaw angle → $\psi$

$e$ → Horizontal offset CG.

$L_{tail}$ → CG, rear overhang dist
$S_{tail}$ → Lateral tail disp.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